

Quant Research

Three Studies in Signal versus Noise

A complete guide to the projects, their methods, their results, and what they mean

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Multi-Period Portfolio Optimization · Eigenportfolios via Random Matrix Theory ·
HMM Regime Detection

Baseline of record: survivor universe, 2005–2024 (P3: 2000–2024) · IS/OOS split at 2018 · All numbers regenerable from a clean clone; last full reproduction 2026-08-23 verified P1 and P2 bit-exact and P3 within its published fragility band.

How to Read This Report

Each project chapter follows the same arc: **the question** the project asks, **the machinery** built to answer it, **the results** exactly as the code prints them, and **the interpretation** — what the numbers mean, why they came out this way, and what an informed skeptic would ask. A glossary at the end defines every technical term used. Nothing here is asserted without a number, and no number appears without its context.

Three conventions govern every result: **(1) IS/OOS separation.** Models are tuned only on the in-sample years (through 2017); the out-of-sample years (2018–2024) are graded, never touched for selection. A written discipline statement with a five-point evidence trail backs this (baselines predate experiments in git history; grids carried pre-stated no-retuning rules; machine-checked leak tests). **(2) Costs everywhere.** Every strategy pays the same transaction-cost model. **(3) Bands over points where fragility is measured.** When a result is sensitive to random seeds, it is quoted as a distribution, not a single lucky draw.

The through-line to hold onto: each project was built to showcase a sophisticated technique, and each concluded — after every correctness fix, against every benchmark, across every sensitivity sweep — that **the naive alternative was at least as good on this data**. Those three honest null results, and the machinery that makes them trustworthy, are the actual contribution.

The Data, Once

Universe: hand-picked 2024-era U.S. large caps traded back to 2005 — deliberately a **survivor universe**, disclosed and measured rather than hidden: portfolios of these names earn roughly **+350–390 basis points per year** over an investable equal-weight benchmark (RSP) purely from the privilege of having picked winners in hindsight. This inflates all performance *levels* roughly equally; *comparisons between strategies* sharing the universe remain meaningful. P1 uses 46 names (after data cleaning), P2 uses 151, P3 uses only index-level series (S&P 500, VIX, Treasury ETF). Prices come from Yahoo Finance, pinned to parquet snapshots; Fama-French factors from Ken French's library, real data enforced by code.

	P1 — Portfolio Optimization	P2 — Eigenportfolios/RMT	P3 — HMM Regimes
Assets	46 large caps	151 large caps	SPY, VIX, TLT
Period	2005–2024	2005–2024	2000–2024
IS / OOS	2005–17 / 2018–24	2005–17 / 2018–24	2000–17 / 2018–24
Rebalance	Monthly	Monthly	Monthly
Core question	Does optimization beat 1/N net of costs?	Does denoising the covariance help?	Can detected regimes be traded?
Honest answer	No — forecasts too weak to justify trading	Along — the tie survived five probes	Detection yes; trading no

Project 1 — Multi-Period Portfolio Optimization

The Question

Classical portfolio theory says: forecast returns, estimate risk, optimize weights. Practitioners object that trading costs eat the theory alive. This project builds the full modern answer to that objection — a **multi-period convex optimizer** whose objective simultaneously weighs expected return (from a rolling Fama-French three-factor model), risk (Ledoit-Wolf shrunk covariance, risk-aversion $\gamma=2$), a turnover regularizer ($\kappa=0.005$), and an explicit transaction-cost model (10 bps proportional + a 3/2-power market-impact term) — and asks whether it beats five benchmarks that don't bother: equal weight (1/N), classical single-period mean-variance, minimum variance, risk parity, and buy-and-hold S&P 500.

The Machinery

Every month for 20 years: re-estimate 46 factor regressions on the trailing year, rebuild the covariance, solve the constrained program (long-only, 5% position cap, 40% turnover cap) with CVXPY, charge realized costs, roll forward. Two engineering guarantees matter: the cost formula the optimizer *penalizes* and the one the backtest *charges* are the same function in the same file (machine-checked on random trades to 10 decimal places), and every one of the 227 monthly solves is verified optimal — zero silent fallbacks.

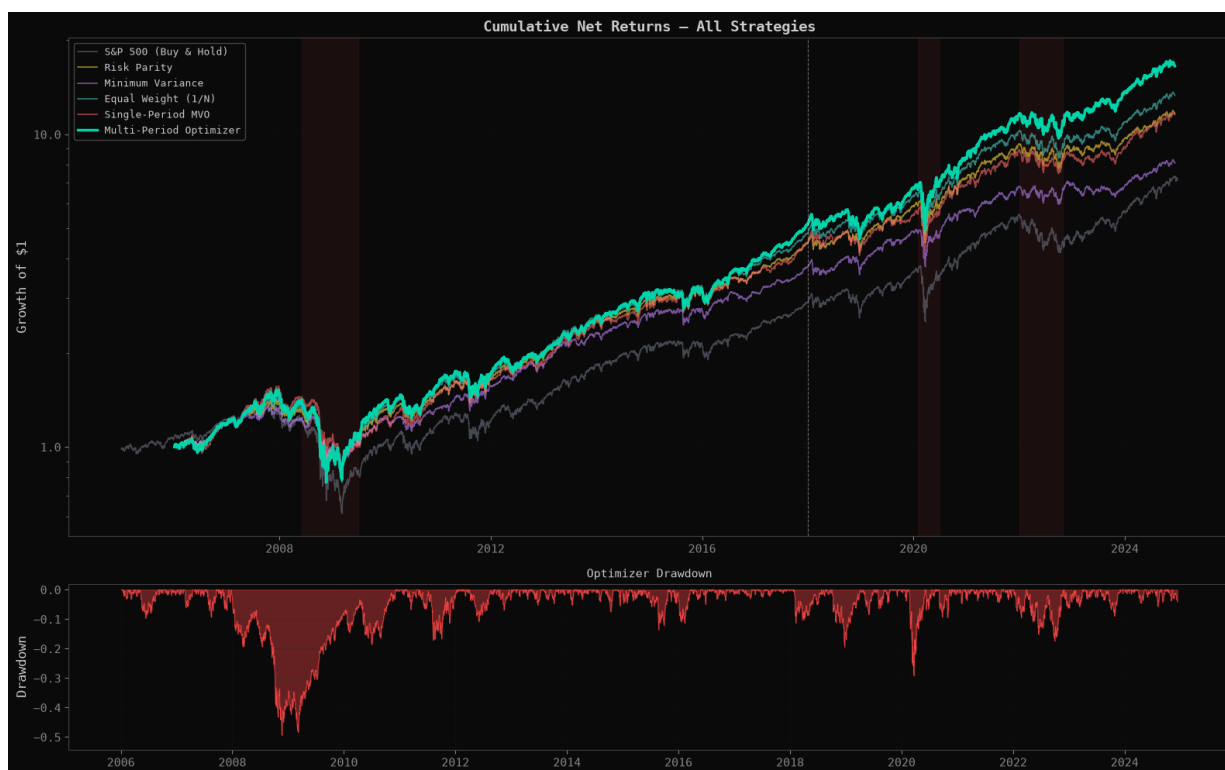
Results

```
PERFORMANCE — OOS (2018-2024), net of costs
Strategy      Return    Vol    Sharpe    SR 95% CI    MaxDD
Multi-Period Optimizer  18.70%  19.15%  0.846    [-0.02, 1.71] -29.21%
Single-Period MVO      15.17%  17.55%  0.722    [-0.11, 1.56] -27.56%
Equal Weight (1/N)     16.27%  17.53%  0.785    [-0.07, 1.64] -27.76%
Minimum Variance       12.17%  14.59%  0.663    [-0.16, 1.48] -24.12%
Risk Parity           15.16%  16.30%  0.777    [-0.07, 1.63] -26.39%
S&P 500 (Buy & Hold)   14.84%  19.46%  0.634    [-0.18, 1.45] -33.72%

Turnover & costs: Optimizer 0.0000 / 0.0 bps    MVO 0.3281 / 825 bps
                  MinVar 0.1195 / 289 bps      RiskParity 0.0329 / 76 bps
Solver health: 227/227 optimal, zero equal-weight fallbacks (all strategies)

IS (2005-2017): Optimizer 0.690 | MVO 0.682 | EW 0.685 | MinVar 0.680
                  RiskParity 0.693 | S&P 0.395
```

Look at the turnover row before the Sharpe column: **the table-topping optimizer never traded**. Its 0.846 Sharpe belongs to a portfolio frozen at its very first solve in 2006 — nineteen years of monthly re-optimization in which the machine concluded, every single month, that no trade cleared the hurdle. Whether that is a bug or an answer is exactly what the sensitivity grid settles.

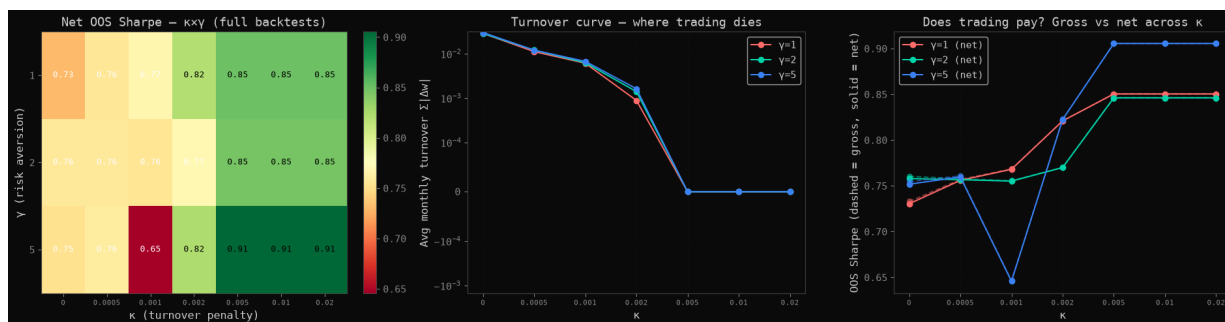


Cumulative net returns, all strategies, log scale. The IS/OOS boundary is the dashed line at 2018.

The $\kappa \times \gamma$ Grid — Why the Freeze Is the Answer

A 21-cell grid (κ from 0 to $4 \times$ the baseline, log-spaced $\times \gamma \in \{1, 2, 5\}$), each cell a complete 227-rebalance backtest, published with a pre-stated no-retuning rule (baseline stays $\kappa=0.005$; anything else is an in-sample statement only). Three findings:

- 1. The freeze boundary sits exactly at the baseline.** At every γ tested, the optimizer trades below $\kappa=0.005$ and freezes at or above it — the hand-chosen "reasonable" penalty landed precisely on the trade/no-trade cliff.
- 2. Costs barely bite.** Across all twelve trading cells, the gap between gross and net Sharpe never exceeds 0.003 — at realized turnover, the much-feared transaction costs are nearly irrelevant.
- 3. Trading still doesn't pay.** Only 2 of 12 trading cells beat equal weight net OOS, both within noise; and the cell in-sample selection would have picked ($\gamma=1$, $\kappa=2\%$) is itself a frozen cell — the in-sample data independently concludes that trading isn't worth it.



The trade-off surface: net OOS Sharpe heatmap; the turnover curve showing where trading dies; gross vs net across κ .

Interpretation

Costs are not the binding constraint — forecast quality is. Trailing-average Fama-French forecasts simply do not contain enough signal to justify moving money monthly against even a modest hurdle. The optimizer, given honest inputs and honest costs, correctly chose restraint; its "degenerate" buy-and-hold behavior is the optimal policy for its information set. This independently reproduces the most famous null result in portfolio choice (DeMiguel, Garlappi & Uppal 2009: naive 1/N is extraordinarily hard to beat) — reproduced here not by assumption but by a machine built and tuned to refute it. Two caveats keep the claim honest: the optimizer's edge over equal weight (0.846 vs 0.785) has widely overlapping confidence intervals — "suggestive, never established" — and every level in the table is inflated by the survivor universe. The defensible one-liner: *"my optimizer's best trade was refusing to trade, and I have a 21-cell surface proving that's the right answer, not a broken one."*

Project 2 — Eigenportfolios via Random Matrix Theory

The Question

A covariance matrix of 151 assets has 11,476 parameters, estimated from only 252 daily observations per window — roughly 46 parameters per data point. Most of what a sample correlation matrix "sees" is therefore noise. Random Matrix Theory offers a scalpel: the **Marchenko-Pastur law** predicts exactly what the eigenvalue spectrum of pure noise looks like, so anything above its upper edge $\lambda_{+} = \sigma^2(1+\sqrt{q})^2$ is real structure and everything below can be flattened. The question: does this principled denoising build better minimum-variance portfolios than the raw sample matrix — or than Ledoit-Wolf shrinkage, the practitioner's default fix?

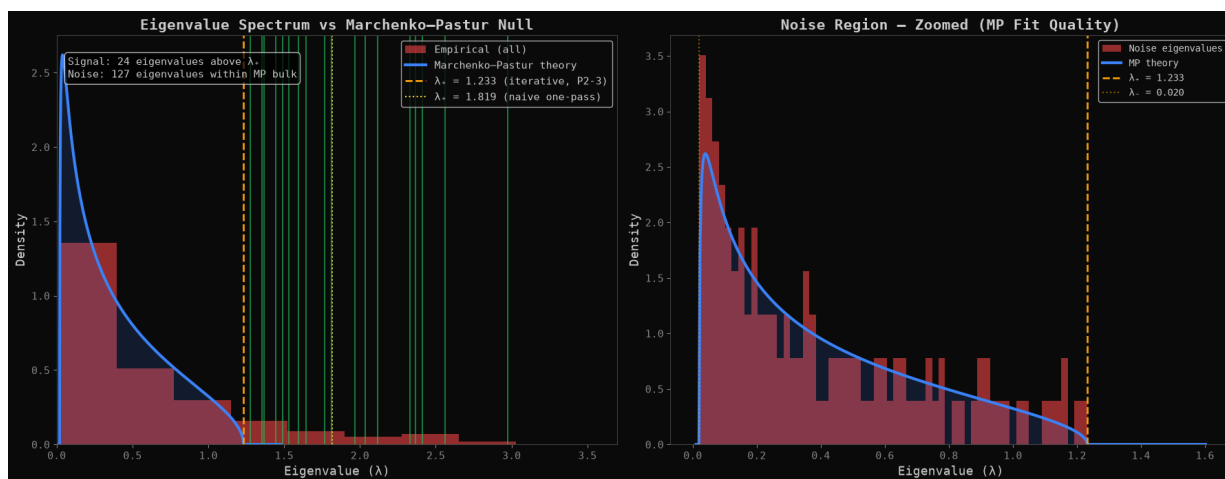
The Machinery

Hand-rolled throughout: eigendecomposition of the correlation matrix each window; an **iterative noise-level estimate** (signal eigenvalues absorb part of the fixed variance budget, so the noise floor σ^2 must be re-estimated excluding them — classify, subtract, lower the fence, repeat until stable); three cleaning variants; then the crucial handoff — reset the cleaned correlation's diagonal to exactly 1 and rescale by each asset's volatility so the optimizer sees a true *covariance* (a min-variance portfolio fed a correlation matrix is blind to volatility, a bug this project found and fixed in itself). Ledoit-Wolf fits raw returns; the raw sample matrix competes as-is.

Results — Structure

```
Marchenko-Pastur fit (last 252-day window, 151 assets, q = 0.599)
sigma^2 (iterative) = 0.392      lambda_+ = 1.2325 (one-pass would say 1.8188)
Signal eigenvalues: 24 of 151    signal variance 67.1% / noise 32.9%
lambda_1 = 30.09 (19.9% of variance, the market) lambda_2 = 14.83 (semis vs
utilities/staples) lambda_3 = 9.24 ... lambda_24 = 1.28
Condition number: raw 1435 -> hard-threshold cleaned 77 (18x more stable)
```

The iterative edge correction matters scientifically: the naive one-pass fence (1.82) hides ten real factors that the corrected fence (1.23) recovers — 24 genuine dimensions of structure, not 14. Eigenportfolio 2 is readable economics: long semiconductors (AMAT, NVDA, QCOM, MU, AMD), short utilities and staples (SO, AEP, DUK, WEC, KO) — a growth-versus-defensive axis no one programmed in. And 84% of the correlation matrix's eigenvalues are statistically indistinguishable from noise: the dimensionality problem is real.

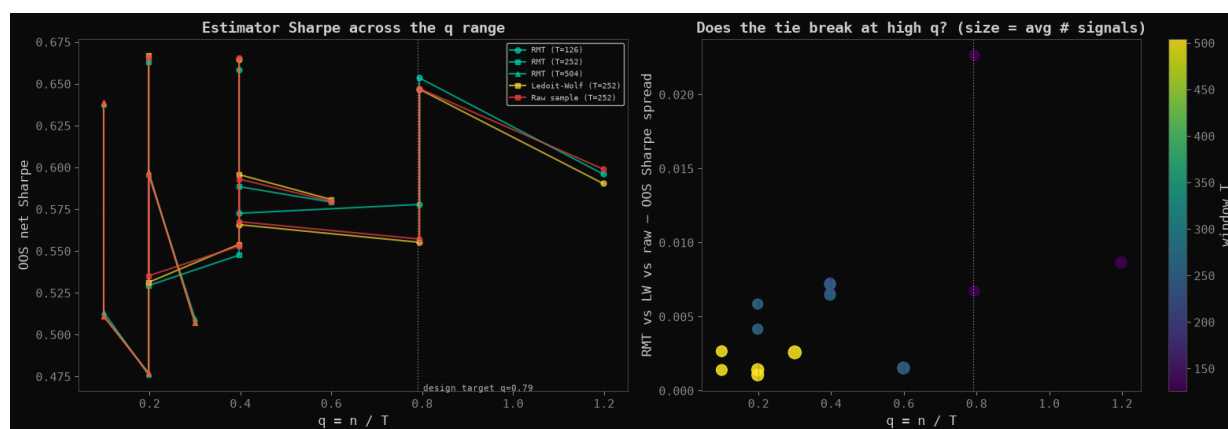


The signature plot: empirical eigenvalue spectrum against the Marchenko-Pastur null, both fences marked.

Results — Portfolios

PERFORMANCE — OOS (2018-2024), net of costs					
Strategy	Return	Vol	Sharpe	MaxDD	Non-opt
RMT Hard Threshold	10.79%	14.31%	0.579	-28.37%	0/217
RMT Targeted Shrinkage	10.79%	14.32%	0.579	-28.46%	0/217
RMT Constant Residual	10.79%	14.31%	0.579	-28.37%	0/217
Ledoit-Wolf Min-Var	10.81%	14.31%	0.581	-28.50%	0/217
Raw Sample Min-Var	10.80%	14.32%	0.580	-28.59%	0/217
Equal Weight (1/N)	15.14%	18.15%	0.697	-31.04%	
S&P 500 (Buy & Hold)	14.84%	19.46%	0.634	-33.72%	

Read the middle five rows: RMT, Ledoit-Wolf, and the raw sample matrix are **indistinguishable — a Sharpe spread of 0.002**. All five do deliver what min-variance promises (lowest volatility, shallowest drawdowns), but the denoising sophistication separates nothing. That tie was then attacked five separate ways: near-death survivor cohort (spread 0.006), smooth cohort (0.002), six random sub-universes (≤ 0.009), and a 15-cell window $\times$ universe grid marching q from 0.10 to 1.20 — **visiting the design-target $q=0.794$ exactly** (max spread of the whole study, 0.023, occurs there; its twin seed shows 0.007) and the formally singular $q=1.198$ corner (spread 0.009, zero solver failures in 669 solves). Five probes, zero separations, question pre-stated in the notebook before any number existed.



The q surface: estimator Sharpes and their spread across the full $q = n/T$ range; design target marked.

Interpretation

Why doesn't cleaning matter here? Min-variance weights are dominated by the top eigenvectors — the market factor and the big sector axes — and all three estimators agree almost perfectly about those. Their disagreements live in the noise bulk, which the optimizer barely touches once a PSD floor prevents outright explosion. Denoising fixes a numerical problem (condition number 1435 $\rightarrow$ 77) that the portfolio construction was only mildly exposed to. **The contribution is therefore the structural work** — the corrected factor count, the spectrum itself, the crisis-time dimensionality collapse — and a defensible negative: "on this universe and period, RMT cleaning, Ledoit-Wolf, and doing nothing produce the same min-var portfolio, and here are five sweeps that tried to make them differ." Practitioners trust that sentence precisely because the project refused to flatter the method it is named after.

Project 3 — HMM Regime Detection

The Question

Markets seem to have moods — long calm bull stretches, violent crisis clusters. A **hidden Markov model** formalizes this: an unobserved "regime" switches with fixed probabilities, and each regime emits returns with its own mean and volatility. Three questions in one: can the model *detect* regime changes fast enough to matter, can a defensive-switching strategy *trade* them profitably against buy-and-hold and a 60/40 portfolio, and can its regime-aware volatility forecasts beat causal benchmarks (rolling vol, EWMA, the VIX)?

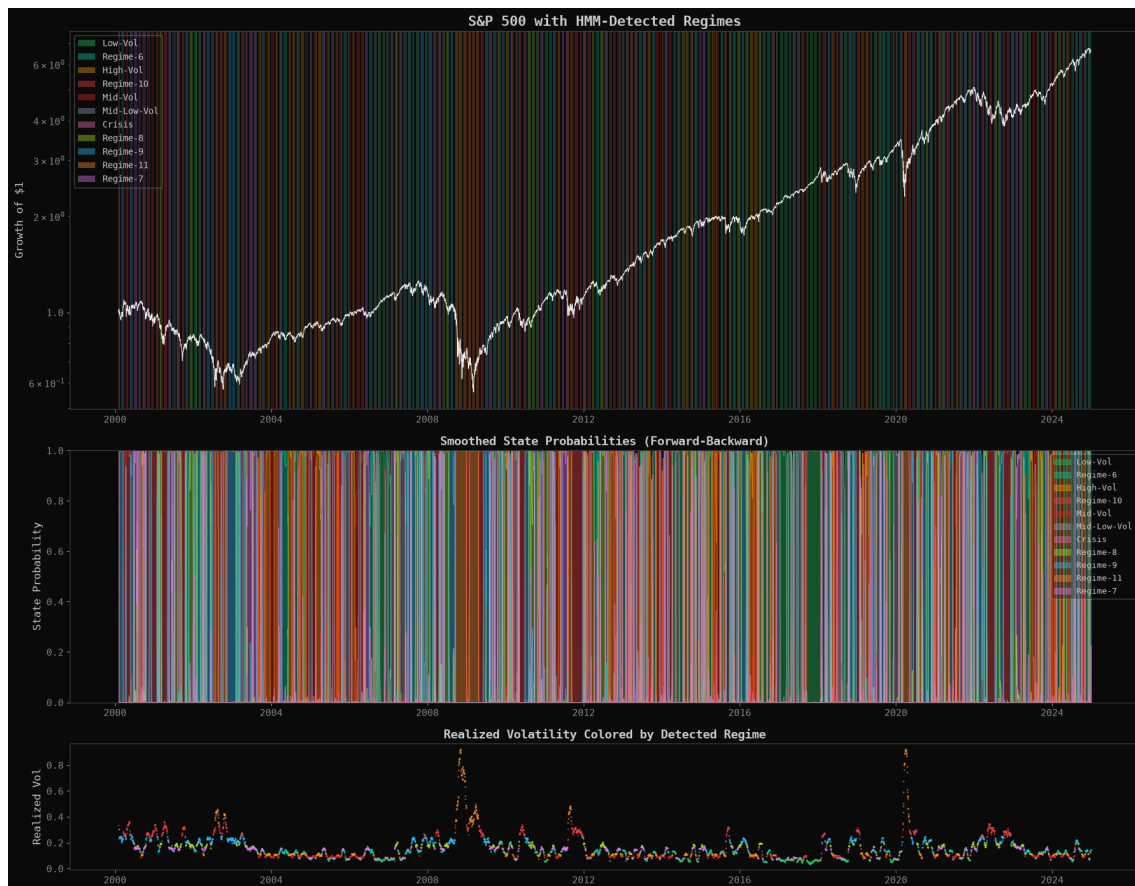
The Machinery

A Gaussian HMM on three daily features (S&P return, 21-day realized volatility, VIX change), refit every month on a trailing six-year window in a strict walk-forward. The anti-lookahead engineering is the deepest in the repo: features are winsorized with **causal expanding-window quantiles** and scaled by frozen in-sample statistics; each backtest window standardizes by its own trailing stats; forecasts are graded on **filtered** probabilities (a hand-rolled log-space forward pass, validated to exact agreement with the reference library at the one time point where filtering and smoothing provably coincide); machine-checked "delete-the-future" tests pin all of it. Because EM fitting is seed-sensitive, the strategy is quoted as a **band across 8 seeds + 3 perturbations**, never a single run.

Results

PERFORMANCE (net) — IS (2000-2017) and OOS (2018-2024)				
	IS Sharpe	OOS Sharpe	OOS Return	OOS MaxDD
HMM Regime Switching	0.740	0.479	8.32%	-29.70%
Buy & Hold SPY	0.481	0.640	14.95%	-33.72%
60/40 Portfolio	0.728	0.482	8.44%	-27.24%
VIX-Based Switching	0.690	0.199	5.74%	-39.76%
Rolling Vol Switching	0.738	0.256	6.62%	-36.59%

FRAGILITY BAND (11 runs): OOS Sharpe 0.406 +/- 0.064, range [0.28, 0.48]
Detection latency: GFC +4 days | COVID +11 days | 2022 grind: missed
Crisis alpha vs B&H: GFC +41.8% | COVID +15.7% | 2022 -7.9%
Vol forecast OOS RMSE: VIX 0.0965 (best) | EWMA 0.1038 | HMM 0.1005
Model selection: corrected BIC falls monotonically through K=12 (never turns);
strategy uses pre-stated 95%-improvement elbow, K=11 (labeled pragmatic)



The signature plot: detected regimes shading twenty-five years of the S&P; 500.

Interpretation — Four Layers

1. The detector works. Regimes align with every named market era; the 2008 crash is flagged within 4 trading days and COVID within 11 — fast for a realized-vol instrument. The model is a good thermometer. **2. The strategy loses, and the eras explain why.** In-sample (2000–2017, two historic crashes) regime-switching *beat* buy-and-hold 0.740 to 0.481; out-of-sample (a bull decade with V-shaped dips) the ranking inverted, 0.41 ± 0.06 vs 0.640. Regime switching is crash insurance: it earned +42% over the market during the GFC and +16% during COVID, then paid the premium back waiting for crashes that resolved too quickly (2022's slow grind, which realized-vol features never flagged, cost another –8%). Insurance loses money in good decades — that is not a modeling failure, it is what insurance is. **3. The band is the honest quote.** The single-run Sharpe (0.479) sits at the very top of its own seed band [0.28, 0.48] — citing the point would overstate the strategy by luck; the entire band sits below buy-and-hold, making the conclusion seed-independent. A fresh 2026-08 reproduction moved these EM-fragile numbers by at most 0.001–0.002 — inside the band, exactly as designed. **4. The regime count is unanswerable as posed.** Corrected BIC decreases monotonically through $K=12$: on fat-tailed daily data, Gaussian mixtures always want more components, so "how many regimes does the market have?" has no finite answer under Gaussian emissions. The strategy's $K=11$ is a pre-stated elbow rule, explicitly labeled pragmatic; a credible count requires heavier-tailed (Student-t) emissions — the designated Phase 4 extension. The forecasting verdict completes the honesty: the VIX — the market's own implied forecast — beats the HMM causally, and hindsight (smoothed) grading would have flattered the HMM by only ~ 0.001 RMSE, a gap this project measured rather than hid.

Synthesis — What the Three Projects Say Together

One sentence each: **P1** — weak forecasts, not transaction costs, are what kills active rebalancing. **P2** — covariance denoising elegance doesn't move portfolios when the dominant factors are already well-estimated. **P3** — detecting market states and profiting from them are different problems, separated by latency and insurance economics.

Together: on twenty years of large-cap data, **the naive benchmarks — equal weight and buy-and-hold — beat every sophisticated method this repo built to outdo them.** Each null is earned, not assumed: pre-stated questions, published sensitivity surfaces (21-cell $\kappa \times \gamma$, 15-cell window $\times$ universe, 11-run fragility band), and a written no-peeking statement with a five-point evidence trail. The repo also caught and fixed twenty audited defects along the way — including a return-aggregation bug whose fix *reversed* a headline ranking, and a BIC mis-scaling found mid-project and adopted into the audit rather than quietly patched. Every fix has a before/after entry in the changelog.

Limitations, Disclosed

Survivorship (the big one): the universe is 2024's winners traded from 2005 — measured at $\approx +350$ –390 bp/yr of level inflation vs RSP, stable across sub-periods; the dead of 2008 are censored, so crisis drawdowns are understated and min-var's GFC resilience is flattered. This is the study's fixed, documented scope; a CRSP point-in-time version is a designated future extension, not a re-baseline. **EM fragility:** P3's strategy Sharpe varies ± 0.06 across seeds — hence bands. **Statistical power:** seven OOS years put ± 0.8 -wide confidence intervals on every Sharpe; ranking claims are suggestive, not established. **Remaining known gaps** (owned, scheduled): full Baum-Welch hand-implementation (the causal forward pass is already built and validated), a true GARCH(1,1) benchmark, Student-t emissions, and cost amortization timing.

Reproducibility — Tested Under Fire

Environment pinned in requirements.lock; data pinned in parquet snapshots; 57 automated tests including cost-contract and delete-the-future causality invariants; every figure and table in this report regenerates from a clean clone with one command per notebook. This is not theoretical: in August 2026 a folder rename silently broke the virtual environment, three notebook runs no-op'd, the staleness discipline (write-proof + timestamp + freshness print) caught all three false successes, the environment was rebuilt from the lock file in two minutes, and the full re-run reproduced P1 and P2 to the last printed digit and P3 inside its published band. The reproducibility story has a passing live-fire test.

Glossary — Every Term, Plainly

Sharpe ratio. Return earned per unit of risk taken: (annualized return – risk-free) / annualized volatility. 0.5 is respectable, 1.0 is strong. Its error bar shrinks only with years of data — 7 OOS years gives roughly ± 0.8 , which is why overlapping intervals mean "not established."

IS / OOS. In-sample (data you tuned on) vs out-of-sample (data you only graded on). The entire value of an OOS number rests on never having used it for selection — hence the written discipline statement.

Walk-forward backtest. Simulate decisions date by date, letting each decision see only data before that date. The machine-checkable definition: any quantity known at time t must be unchanged if all post- t data is deleted.

Simple vs log returns. Percent change vs its natural log. Logs add across time; percents average across assets. Portfolio return = weighted average of simple returns, always — averaging log returns instead silently taxes diversified portfolios (a bug this repo found, fixed, and measured: it had reversed P2's headline ranking).

Equal weight (1/N). Put $1/N$ of capital in each asset, rebalance mechanically. The humble benchmark that all three projects failed to beat — consistent with a large academic literature.

Ledoit-Wolf shrinkage. Blend the noisy sample covariance toward a simple target with an analytically optimal weight. The practitioner's standard cure for high-dimensional noise.

$q = n/T$. Assets per observation — the starvation ratio of covariance estimation. At q near 1 the sample matrix is mostly noise; above 1 it is singular.

Marchenko-Pastur law. The exact eigenvalue distribution pure noise would produce. Eigenvalues above its edge $\lambda_{\max} = \sigma^2(1 + \sqrt{q})^2$ carry real structure; the edge must be estimated iteratively because signal eigenvalues consume part of the variance budget.

Eigenportfolio. A portfolio along one eigenvector of the correlation matrix. The first is "the market"; later ones are statistical sector/style contrasts the data volunteers on its own.

Condition number. Largest $\div$ smallest eigenvalue — how explosively a matrix inverts. Denoising cut it 1435 $\rightarrow$ 77 here.

Minimum-variance portfolio. The weights minimizing portfolio variance (closed form $\Sigma^{-1} / (1^T \Sigma^{-1} 1)$ unconstrained). Its job is low volatility, not high Sharpe — judged accordingly.

Transaction-cost model. Charge per trade: proportional (spread/commission, 10 bps) plus $3/2$ -power market impact (big orders move prices). Here the optimizer's penalty and the backtest's charge are one function, tested equal.

Turnover / κ . $\Sigma |\Delta w|$ per rebalance — how much of the portfolio moves. κ is a deliberate extra penalty on it (a regularizer against over-trusting forecasts), distinct from real costs.

Hidden Markov model. Observed data generated by an unobserved state that switches with fixed probabilities; fit by the EM (Baum-Welch) algorithm, which is seed-sensitive — hence bands.

Filtered vs smoothed. Filtered probability uses data up to now (tradeable); smoothed uses the whole history including the future (hindsight — plots only). They coincide only at the final time step, an identity this repo exploits both to trade legally and to validate its hand-rolled forward pass.

BIC. A model-selection score: fit minus a complexity penalty. On fat-tailed daily data with Gaussian emissions it never stops rewarding extra regimes — a finding here, not a failure.

Winsorization. Clipping extreme values at chosen quantiles. Done causally here: each day's clip bounds computed only from data up to that day.

Survivorship bias. Backtesting today's winners through history. Two faces: selection (you picked names that later won — measured at +350–390 bp/yr here) and censoring (the dead are missing — unmeasurable without point-in-time data, disclosed).

Fragility band. The distribution of a result across random seeds and small perturbations. Cite the band; a single run may sit — as it literally did here — at the band's maximum.

Pre-stated question / no-retuning rule. Writing what would count as an answer into the notebook before the numbers run — the research analog of pre-registration, done for both sensitivity grids in this repo.

Where Everything Lives

results/tables/ — the printed metric tables · results/figures/ — all 28 figures · results/CHANGELOG.md — before/after for every fix that moved a number · docs/LIMITATIONS.md — measured limitations · results/OOS_DISCIPLINE.md — the no-peeking statement and evidence trail · docs/DECISION_LOG.md — one line per working session · 57 tests via pytest · regenerate any project: jupyter nbconvert --to notebook --execute --inplace projects/<p>/notebooks/exploration.ipynb